

- ii Task prioritization, due dates, and status tracking
- iii Automated workflow routing, reminders, and escalation mechanisms.
- iv Real-time task dashboards for individuals and leaders.

E. Paper Preparation & Multi-level Approval Workflows

- i Collaborative drafting environment with track changes and version comparison
- ii Multi-stage approval workflow including but not limited to:
Member → Group Leader → Delegation Leader → IC Division → CS/NA/CSS → DG → MoPSW (where applicable)
- iii Locking and finalization of approved documents to prevent unauthorized edits.

F. Correspondence Group Management

- i Nomination and management of Correspondence Group members.
- ii Upload and review of correspondence group tasks, inputs, and progress updates.
- iii Group-level consolidation and reporting mechanisms.

G. Meeting-time Collaboration & Discussion Interface

- i Controlled real-time collaboration interface during live meetings.
- ii Discussion boards for agenda-wise deliberations and last-minute changes.
- iii Role-based editing and commenting permissions during meetings.

H. Dashboards, Reports & Analytics

- i Role-based dashboards (Admin, Delegation Leader, Coordinator, Member).
- ii Committee-wise, meeting-wise, and agenda-wise dashboards.
- iii Tracking of tasks, submissions, pending approvals, and outcomes.
- iv Analytical insights on participation, performance, and timelines.
- v Downloadable and configurable reports in XML, Excel, and PDF formats.

I. Calendar, Alerts & Notifications

- i Integrated calendar displaying upcoming meetings, deadlines, and milestones.
- ii Automated alerts and notifications for agenda updates, task assignments, approvals, and meetings.
- iii Notification delivery through in-portal alerts and email.

J. User Management & Role-Based Access Control

- i Centralized administrative console for user and role management.
- ii Assignment of users to committees, working groups, and roles.
- iii Role-based access enforcement across all modules and data.
- iv Comprehensive access logs and audit trails for all user actions.

K. Compliance & Quality Requirements

- i The application shall strictly adhere to the approved FRS/SRS and all applicable Government of India usability, accessibility, security, and performance standards.
- ii Any deviations shall require prior written approval from DGS.
- iii The SI shall ensure that the solution is extensible, maintainable, and future-ready.

Detailed Functional Requirements are mentioned in Annexure 2

3.4 Technology Stack

The System Integrator (SI) shall propose, implement, and maintain an open, scalable, secure, and future-ready technology stack for the IMO Strategic Digital Platform. The proposed stack shall be subject to approval by the Directorate General of Shipping (DGS) and shall comply with Government of India digital, security, accessibility, and interoperability standards.

The technology stack shall be based on open standards and industry-accepted frameworks, ensuring interoperability, ease of maintenance, and long-term sustainability. The architecture shall support horizontal and vertical scalability to handle increased users, data volumes, and future functional expansion. Security shall be embedded across all layers of the application, and the system shall be designed to avoid vendor lock-in, allowing DGS to migrate, extend, or maintain the solution independently in the future.

The frontend shall be a web-based, responsive interface, developed using standard frameworks (such as Angular, React, or equivalent), compatible with all major modern browsers and devices (desktop, tablet, and mobile). The solution shall comply with Government usability and accessibility guidelines, including WCAG 2.1.

The backend shall be developed using enterprise-grade, open frameworks, following a service-oriented or microservices-based architecture. It shall support modular development, stateless services, centralized business logic, workflow management, and role-based access enforcement. All functionalities shall be exposed through well-documented APIs to enable future integrations.

The integration layer shall follow an API-first approach using RESTful services and open data formats. Secure API management mechanisms shall be implemented for authentication, authorization, monitoring, and future extensibility. The data layer shall use enterprise-grade databases suitable for both structured and unstructured data, with appropriate indexing, encryption, automated backups, and recovery mechanisms to ensure data integrity and availability. The platform shall be hosted exclusively on a MeitY-empowered Cloud Service Provider, with appropriate cloud services for high availability, scalability, backup, disaster recovery, and monitoring. Cloud infrastructure and application security controls shall comply with ISO 27001, ISO 27017, ISO 27018, and CERT-In guidelines. Industry-standard tools shall be used for development, testing, version control, and CI/CD, with separate environments for development, testing/UAT, production, and disaster recovery.

The entire source code and related artefacts shall be the exclusive property of DGS. The SI shall ensure that no proprietary dependencies are introduced that restrict future modification or transfer of the system. All third-party components used shall be properly licensed and disclosed.

Any deviation or change in the approved technology stack shall require prior written approval from DGS, and non-compliance with approved standards shall be treated as a contractual breach.

3.5 Cloud Hosting Infrastructure

The System Integrator (SI) shall host the IMO Strategic Digital Platform exclusively on a MeitY-empowered Cloud Service Provider (CSP), in strict compliance with MeitY Guidelines for

Procurement and Use of Cloud Services (GI Cloud / MeghRaj) and other applicable Government of India directives. Use of any non-empanelled cloud infrastructure shall not be permitted at any stage of the project.

The SI shall provision and maintain logically and securely segregated cloud environments for Development, User Acceptance Testing (UAT), Production, and Disaster Recovery (DR). The Production environment shall be designed for high availability, while the DR environment shall ensure business continuity in the event of system failure, cyber incidents, or natural disasters.

A. Resilience, Backup & Disaster Recovery (RPO/RTO)

The cloud architecture shall meet the following **minimum benchmarks**, aligned with MeitY cloud best practices:

- i. **Recovery Point Objective (RPO):** Maximum data loss shall not exceed 24 hours for Production systems
- ii. **Recovery Time Objective (RTO):** Restoration of Production services shall be completed within 8 hours of a disaster declaration.
- iii. Automated daily backups shall be configured, with secure off-site storage and defined retention policies
- iv. Periodic restoration and DR drills shall be conducted, and reports submitted to DGS.

B. Infrastructure Provisioning & Management

- i. The SI shall configure, manage, and optimize all required cloud components, including:
 - Compute, storage, and database services
 - Secure networking, subnets, firewalls, and load balancers
 - Web Application Firewall (WAF) and security monitoring services
 - Centralized logging, monitoring, and alerting mechanisms
- ii. The infrastructure shall be scalable to handle peak usage during international committee meetings without performance degradation.

C. Security & Compliance

The cloud hosting and application environment shall comply with:

- ISO 27001 – Information Security Management
- ISO/IEC 27017 – Cloud Security Controls
- ISO/IEC 27018 – Protection of Personally Identifiable Information in Cloud
- CERT-In advisories and cyber security guidelines

The SI shall ensure encryption of data at rest and in transit, robust identity and access management, security logging, and prompt remediation of vulnerabilities identified through periodic security audits and VAPT.

D. Cloud SLA & Uptime

- i. The SI shall ensure a **minimum 99% application uptime** on a monthly basis for the Production environment, excluding approved maintenance windows.
- ii. Downtime attributable to the SI or CSP shall attract penalties as defined in the SLA.
- iii. The SI shall remain **fully accountable** to DGS for meeting SLA commitments, irrespective of CSP performance

E. Cost Management & Optimization

The SI shall actively monitor cloud usage and implement cost optimization measures, including right-sizing of resources, avoiding idle infrastructure, and providing periodic cloud utilization and cost reports to DGS.

F. Exit Management & Data Portability

In accordance with MeitY Cloud Guidelines, the SI shall ensure:

- i. **Complete data ownership** rests with DGS at all times
- ii. All application data, metadata, logs, and documents shall be **exportable in open and non-proprietary formats**
- iii. At contract termination or exit, the SI shall support seamless migration of the application and data to another MeitY-empanelled CSP without disruption
- iv. The SI shall provide all necessary technical assistance, documentation, configurations, and handholding during the exit or transition period
- v. No data shall be retained by the SI or CSP after exit, and formal data destruction certification shall be provided

The SI shall remain fully responsible for ensuring performance, security, compliance, continuity, and data portability of the cloud-hosted solution throughout the contract lifecycle

3.6 Security Requirements, Audits and Adherence to Standards

The System Integrator (SI) shall be responsible for ensuring **end-to-end data security, privacy, confidentiality, and cyber resilience** of the IMO Strategic Platform throughout the project lifecycle, including design, development, hosting, operations, and exit. The security framework shall be fully aligned with DGS data and cyber security standards, Government of India policies, and global best practices.

The solution shall comply with all applicable security guidelines issued by MeitY, CERT-In, STQC, and shall adhere to ISO/IEC 27001, ISO/IEC 27017 (Cloud Security), ISO/IEC 27018 (Cloud Privacy), and the Digital Personal Data Protection (DPDP) Act, 2023. All application and infrastructure controls shall conform to MeitY's guidelines for procurement and use of cloud services (GI Cloud/MeghRaj) and Government of India security policy frameworks.

The platform shall be hosted exclusively on a **MeitY-empanelled Cloud Service Provider**, with primary and disaster recovery infrastructure located within India. The cloud environment shall be secured through next-generation firewalls, Web Application Firewall (WAF), anti-DDoS mechanisms, role-based access control, secure network segmentation, encryption of data at rest (minimum AES-256) and in transit (TLS 1.2 or above), centralized audit logging, and continuous monitoring.

At the application level, the SI shall ensure secure coding practices in line with OWASP Top 10, including protection against SQL injection, XSS, CSRF, insecure APIs, and session hijacking. Identity and Access Management (IAM) with strong authentication, multi-factor authentication for privileged users, and strict role-based authorization shall be enforced. Source code repositories, CI/CD pipelines, and deployment environments shall be secured with controlled access and integrity checks.

The SI shall implement comprehensive logging, monitoring, and incident response mechanisms, including 24x7 automated monitoring and integration with a Security Information and Event Management (SIEM) system. Detailed logs of user activity, system events, administrative actions, and API access shall be maintained and retained as per regulatory and DGS requirements. A documented incident response plan shall be established for detection, escalation, containment, and resolution of cyber security incidents, in compliance with CERT-In reporting norms.

The SI shall conduct Security Audits through CERT-In empanelled or STQC-authorized agencies. A minimum of three comprehensive security audits shall be carried out during the project lifecycle:

- (i) Pre-Go-Live Audit prior to production rollout,
- (ii) Mid-Term Audit during the Operations & Maintenance phase, and
- (iii) Final Audit prior to project closure and handover.

All critical and high-severity vulnerabilities identified during audits shall be remediated by the SI at no additional cost, followed by re-testing and submission of a **Final Security Compliance and Clearance Certificate**. The SI shall obtain and maintain **STQC certification** for the project as per defined timelines; failure to do so may result in deferment of payments.

The SI shall ensure **strong privacy and confidentiality controls**, including protection of Personally Identifiable Information (PII), restricted access based on organizational hierarchy, anonymization or pseudonymization of sensitive data where applicable, and consent mechanisms for data usage as required under the DPDPA Act.

Robust backup and disaster recovery mechanisms shall be implemented, including encrypted automated backups, defined **RPO/RTO benchmarks**, periodic DR drills, and documented business continuity procedures to ensure uninterrupted service delivery.

In case of contract termination or exit, the SI shall ensure **secure data migration and exit management**, including transfer of all data, logs, source code, configurations, and documentation to DGS or a designated agency, in open and non-proprietary formats. Complete data sanitization, credential revocation, and certification of data destruction shall be provided, along with a final handover report.

The SI shall ensure adherence to **open standards and industry standards**, including Interoperability Framework for e-Governance (IFEG), MeitY Model RFP templates, GIGW 3.0, UX Design Guidelines for Government Applications, W3C WCAG 2.1, and the Rights of Persons with Disabilities Act, 2016. The application shall be Web 1 compliant, device-agnostic, accessible, user-friendly, and capable of future integration with other Government systems.

The SI shall remain **fully accountable** for security, compliance, and confidentiality of the system and data, irrespective of involvement of any third-party vendors or Cloud Service Providers.

3.7 Testing, Quality Assurance & Acceptance

The System Integrator (SI) shall be responsible for establishing and executing a comprehensive testing and quality assurance framework to ensure that the IMO Strategic Platform meets all functional, technical, performance, and security requirements as approved by DGS.

The SI shall prepare a **detailed Test Strategy and Test Plans**, covering unit testing, functional testing, system integration testing (SIT), performance and load testing, security testing, and user acceptance testing (UAT). All test cases shall be mapped to the approved FRS/SRS through a Requirement Traceability Matrix (RTM).

The SI shall conduct **Functional Testing, SIT, and Performance/Load Testing** to validate system behaviour, workflow integrity, concurrency handling, response times, and stability under peak usage conditions. Any defects identified shall be recorded, tracked, and resolved in accordance with defined severity levels and timelines.

The SI shall provide full support to DGS during **User Acceptance Testing (UAT)**, including preparation of UAT scripts, test data, environment readiness, defect resolution, and re-testing. All observations raised during UAT shall be addressed and formally closed before seeking acceptance.

The SI shall conduct **Security Testing, including Vulnerability Assessment and Penetration Testing (VAPT)**, through a CERT-In empanelled auditor, and shall remediate all identified vulnerabilities. Security clearance and compliance certificates shall be obtained prior to production deployment.

Formal stage-wise acceptance shall be obtained from DGS at each milestone, including completion of development, successful UAT, security clearance, and Go-Live readiness. No phase shall be considered complete without written acceptance from DGS.

3.8 Deployment, Go-Live & Stabilization

The System Integrator (SI) shall be responsible for the deployment of the **IMO Strategic Engagement Platform** in the approved production cloud environment, following successful completion of testing, security audits, and User Acceptance Testing (UAT).

The SI shall support **data migration** from existing systems or repositories, wherever applicable, including data validation and reconciliation to ensure accuracy and completeness. **Prior to Go-Live**, the SI shall perform **system hardening and Go-Live readiness checks**, including verification of security configurations, performance benchmarks, backup and recovery mechanisms, monitoring tools, and operational procedures.

The SI shall provide **comprehensive Go-Live support** and a defined **stabilization period** to ensure smooth transition to production operations. During this phase, the SI shall closely monitor system performance, user issues, and operational incidents, and promptly address any defects or issues encountered.

All Go-Live and post-Go-Live issues shall be resolved within the **agreed Service Level Agreements (SLAs)**. The system shall be formally accepted for operational use only after successful stabilization and written confirmation from DGS.

3.9 Training & Capacity Building

- Design and deliver role-based training programs for administrators and users.
- Conduct onsite/online training sessions for DGS, MMDs, and MoPSW.
- Prepare and supply user manuals, admin manuals, SOPs, FAQs, and training materials.
- Ensure user readiness and adoption prior to and after Go-Live.

3.10 Implementation & Rollout

The System Integrator (SI) shall implement the solution using a **phased or full rollout approach**, as approved by the Directorate General of Shipping (DGS). The rollout strategy shall be finalized in consultation with DGS to ensure readiness of users, infrastructure, and operational processes.

The SI shall support **pilot implementation**, incorporate feedback from pilot users, and refine the system prior to wider deployment. Where required, the SI shall facilitate **organization-wide or nationwide rollout**, covering DGS, MMDs, MoPSW, and other identified stakeholders.

Throughout the rollout phase, the SI shall ensure **minimal disruption to ongoing operations**, provide adequate user support, and address operational issues in a timely manner. Formal rollout completion shall be confirmed through written acceptance from DGS.

3.11 sPost-Go-Live Warranty & Operations and Maintenance

The SI shall provide a **one (1) year comprehensive warranty** from the date of Go-Live, during which all defects, errors, and performance issues shall be rectified at **no additional cost** to DGS. The warranty shall also include minor enhancements, patches, and updates necessary to ensure stable system operation.

Following the warranty period, the SI shall provide two (2) years of **Operations and Maintenance (O&M)** services. O&M shall include 24x7 application support and helpdesk services, incident, problem, and change management, continuous system monitoring, and cloud infrastructure management.

The SI shall ensure **minimum 99% application uptime**, adhere to defined SLAs, and promptly resolve issues within agreed timelines. O&M services shall also include regular enhancements, security patches, version updates, and **quarterly security audits** including VAPT, along with monitoring and optimization of cloud resources.

The selected System Integrator (SI) shall be responsible for end-to-end operation and maintenance (O&M) of the deployed platform and its associated infrastructure for the entire O&M period, ensuring availability, performance, security, and compliance with defined SLAs.

The SI shall provide 24/7 system administration and monitoring of application, database, and cloud infrastructure, including proactive performance monitoring, preventive and reactive maintenance, capacity tuning, and alert-based corrective actions. A minimum 99% uptime of the complete solution stack shall be ensured, with regular review of uptime metrics and mitigation of performance bottlenecks.

A centralized Helpdesk shall be established to log, track, and resolve incidents and service requests through a tiered support mechanism within agreed SLA timelines.

The SI shall implement and maintain security and cyber resilience controls aligned with ISO 27001, ISO 27002, CERT-In guidelines, and DGS cyber security standards, including regular patch management, vulnerability assessments, security audits, antivirus/endpoint protection, and timely response to security advisories.

The SI shall ensure automated and secure data backups, maintain Business Continuity and Disaster Recovery (BCP/DR) arrangements, and restore services as per approved RPO/RTO benchmarks.

Software patches, updates, and upgrades shall be deployed after validation in a staging environment, ensuring zero or minimal disruption, with no unapproved downtime counted against SLAs.

The SI shall maintain Standard Operating Procedures (SOPs) aligned with ITIL/ISO 20000, update them periodically with DGS approval, and conduct knowledge transfer and training for DGS support teams.

Preventive and reactive maintenance shall be carried out, preferably outside business hours. The SI shall ensure availability of all necessary tools and O&M resources at no additional cost during the O&M period.

The SI shall maintain audit logs, archive them securely for the agreed period, share periodic compliance reports with DGS, and coordinate with OEMs or third-party vendors using a defined escalation matrix.

Upgrades, Standards Compliance & Quality Audits

The SI shall provide all updates, patches, and upgrades free of cost during the O&M period and ensure alignment with applicable national and international software versions within agreed timelines. The platform shall comply with prescribed industry and open standards covering workflow, interoperability, security, service management, documentation, and accessibility. The SI shall conduct periodic internal quality audits aligned with CMMI, ITIL, ISO 20000, ISO 27001, etc., and support any audits conducted by DGS or its appointed agencies.

3.12 Integration & Interoperability

The System Integrator (SI) shall ensure that the platform is designed using open standards and interoperable architecture, enabling seamless integration with existing and future systems of DGS, MMDs, MoPSW, and other authorized Government platforms, as and when required.

The system shall support import and export of data and documents in standard and open formats including PDF, Word, Excel, XML, and other formats prescribed by Government guidelines to facilitate information exchange, reporting, and archival.

The SI shall enable third-party integrations, including but not limited to Government portal identity/authentication services, notification services (email/SMS), document repositories, tools, and other authorized external systems, through secure API-based mechanisms. All integrations shall comply with MeitY interoperability standards, security guidelines, and DGS governance policies.

The SI shall ensure that all authentication and identity verification mechanisms comply with 2000, MeitY guidelines, and applicable CERT-In advisories.

3.13 Change Management and Change Requests

The System Integrator (SI) shall be responsible for establishing and implementing a structured Change Management framework to ensure smooth adoption of the platform across all stakeholders, while enabling controlled incorporation of functional, technical, and process-related changes during the project lifecycle.

Changes treated as enhancement or modification services may include, but are not limited to, functional changes in the application, development of new modules, sub-modules, forms or reports, modifications to workflows or the core application framework, integration with additional third-party systems, and augmentation of onsite or offsite technical support resources, as required by DGS.

A. Change Management Strategy

The SI shall design and execute a comprehensive change management plan addressing people, process, and technology aspects. This shall include stakeholder communication, user training, revision of standard operating procedures (SOPs), transition planning, and measures to mitigate resistance to change. The SI shall work closely with DGS to identify potential change areas, assess impacts, and plan phased or controlled rollouts of approved changes.

B. Stakeholder Engagement and Communication

The SI shall ensure timely, transparent, and continuous communication with stakeholders, including officials of DGS, MMDs, MoPSW, and other identified users. The SI shall conduct awareness sessions, periodic briefings, and provide helpdesk support to facilitate adoption of the platform and any approved changes.

C. Capacity Building and Documentation

The SI shall provide structured capacity-building support through virtual and/or physical training sessions, train-the-trainer programs, user manuals, videos, and other learning materials. A centralized knowledge repository shall be maintained, capturing version histories, FAQs, SOPs, and documentation related to all implemented changes.

D. Change Request (CR) Process

Any Change Request (CR) shall be initiated formally in writing by the Competent Authority of DGS. Upon receipt of a CR, the SI shall, within the timeframe specified by DGS, submit a detailed response covering:

- Technical feasibility of the proposed change
- Effort estimation in terms of resources and work units
- Financial implications, if any
- Proposed implementation timeline and schedule

DGS shall review the submission and, upon acceptance, issue a formal written approval authorizing implementation. The SI shall commence work on the CR only after receiving such approval and shall raise invoices strictly as per the agreed contractual terms.

E. Scope Creep Allowance

The SI shall accommodate cumulative approved Change Requests up to 20% of the original Contract Value, covering both functional and technical enhancements. Any Change Requests exceeding this threshold shall be subject to a separate commercial proposal and contractual approval by DGS.

3.14 Project Governance and Management

The System Integrator (SI) shall be responsible for effective project governance, management, monitoring, and control throughout the project lifecycle to ensure timely, quality-driven, and risk-mitigated delivery of the solution.

The SI shall prepare, maintain, and periodically update a comprehensive Project Management Plan (PMP) covering scope, timelines, milestones, resources, risks, quality, communication, change management, and testing strategies. The project shall be executed strictly in accordance with the approved plan.

The SI shall conduct a formal project kick-off meeting and establish a clear governance structure, defining roles, responsibilities, escalation mechanisms, and decision-making authority in consultation with DGS. A structured communication plan shall be developed with defined reporting frequencies, review meetings, and coordination mechanisms.

The SI shall deploy named and qualified key resources from its own payroll for the project and ensure their continuous availability for coordination with DGS. Performance of key personnel shall be reviewed on a monthly basis, and any replacement shall be carried out in a timely manner with personnel of equivalent qualifications and experience, ensuring adequate overlap for knowledge continuity. The SI shall proactively manage attrition and report causes of any high attrition along with corrective measures.

Project execution shall be continuously monitored using a Project Management Information System (PMIS). The SI shall submit regular status reports (monthly or as required), highlighting progress against milestones, risks, issues, deviations, and mitigation actions. All changes shall be managed strictly through the approved change control process.

During the Operations & Maintenance phase, the SI shall continue governance and monitoring activities, ensuring 24x7 system monitoring, SLA compliance, periodic updates, security management, and helpdesk support for the agreed contract period.

3.15 Exit Management Plan

The SI shall prepare and maintain a comprehensive Exit Management Plan from project inception to ensure a smooth, secure, and disruption-free transition of the system, data, and services to DGS or a new vendor at contract completion or termination.

The SI shall provide full cooperation during the exit process, including detailed knowledge transfer and handover. This shall include submission of complete and updated documentation such as FRS, SRS, HLD, LLD, test plans and reports, user and admin manuals, security and cloud architecture documents, and as-built documentation.

The SI shall conduct structured knowledge transfer sessions for DGS IT and functional teams and shall hand over the complete source code, scripts, configurations, deployment artefacts, and credentials (after security sanitization). The SI shall ensure continuity of services during the transition period and certify that no data or system access is retained post-exit.

3.16 Security, Integrity & Confidentiality

- A. **Web Services Security:** System shall comply with all the web services including routing, management, publication, and discovery should be carried out in a secure manner. Those who are using the web services should be able to utilize security services such as authentication, authorization, encryption, and auditing. Encryption of data shall take place at client level itself. Application server shall be enabled with SSL security.
- B. **Data Integrity and Confidentiality:** Data integrity techniques need to be deployed to ensure that information has not been altered or modified during transmission without detection. Similarly, data confidentiality features are also to be applied to ensure that the data is only accessible by the intended parties.
- C. **Transactions and Communications:** With respect to the data transactions and communications, system needs to ensure that the business process is done properly, and the flow of operations is executed in a proper manner.
- D. **Non-Repudiation Security:** The application shall have the non-repudiation security services to protect a party to a transaction against false denial of the occurrence of that transaction by another party. End-to-End integrity and confidentiality of messages must be ensured even in the presence of intermediaries.
- E. **Database Controls:** The database controls for online transaction processing systems like access to database directly, access to database through application, access to log files, access by the remote terminals, DBA controls, backup policy and backup procedures.
- F. **DPDP Act Compliance:** The SI shall implement privacy-by-design controls aligned to the Digital Personal Data Protection Act, 2023, including purpose limitation, consent management, data minimization, retention controls, breach notification protocols, and Data Protection Impact Assessment (DPIA) where applicable.
- G. **Restricted Vendor Access:** Access to digital records (sessions, logs, recordings, assessments) for cloud-based orals shall be strictly limited to named, authorized personnel of the SI, on a "least privilege" basis, enforced by RBAC/MFA. The SI shall maintain immutable access logs and furnish them to DGS on request.
- H. **Encryption & Audit Trails:** Encryption at rest (AES-256) and in transit (TLS 1.2+) is mandatory. Comprehensive audit trails shall capture user identities, timestamps, actions, data objects touched, and IP/device attributes, retained per DGS policy.

4. Expected Project Timeline & Deliverables Schedule

12 Months – Implementation + 3 Years – Operation & Maintenance (including 1 Year Warranty)

(T = 0 denotes the Date of Signing of Contract)

The System Integrator (SI) shall deliver the following deliverables to DGS as part of its obligations under the payment milestones and applicable SLAs. The deliverables listed below are indicative and not exhaustive. The SI shall also deliver all other outputs specified in this RFP or as may be required by DGS during the course of the project. Any conflict with respect to project and/or deliverable timelines shall be resolved by the SI in consultation with DGS and approved by DGS. The approved timelines shall thereafter be binding on the SI.